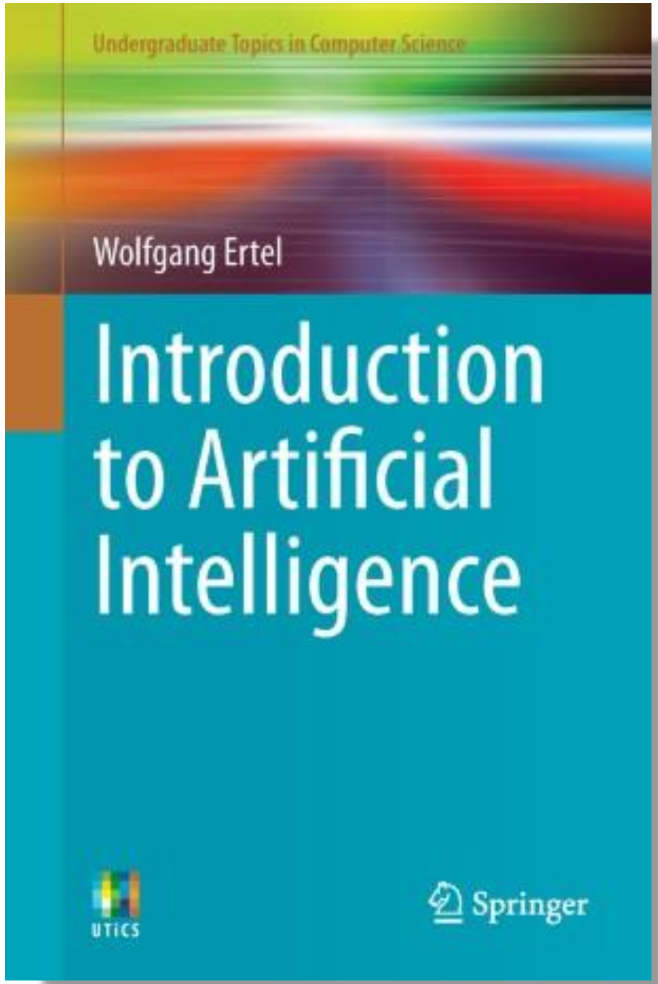
AI Fundamentals

Chapter 1 - Introduction and Examples



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Chapter 1: Introduction and Examples

- We would like to thank the following people for their contribution:
    - Stefaan Haspeslagh, VIVES
    - Yves Sagaert, VIVES
    - Andy Louwyck, VIVES
    - Prof. Danny De Schreye, KU Leuven

AI Fundamentals

Chapter 1: Introduction and Examples

- A mix of theory and practice
- Written examination
- eBook:
  - Introduction to Artifical Intelligence, second edition (ISBN 978-3-319-58486-7)
  - Wolfgang Ertel
  - URL: https://link.springer.com/book/10.1007/978-3-319-58487-4

Contents

About intelligence...

- When should we consider a program *intelligent*?
- When do we consider a creative activity of humans to require *intelligence*?
- Default answers: *never? always?*

Does numeric computation require intelligence?

- For humans?

```
      3 2 3
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      2 8 6 9
  ×     4 5 1
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1 2,9 3,9 1 9
```

- For computers?

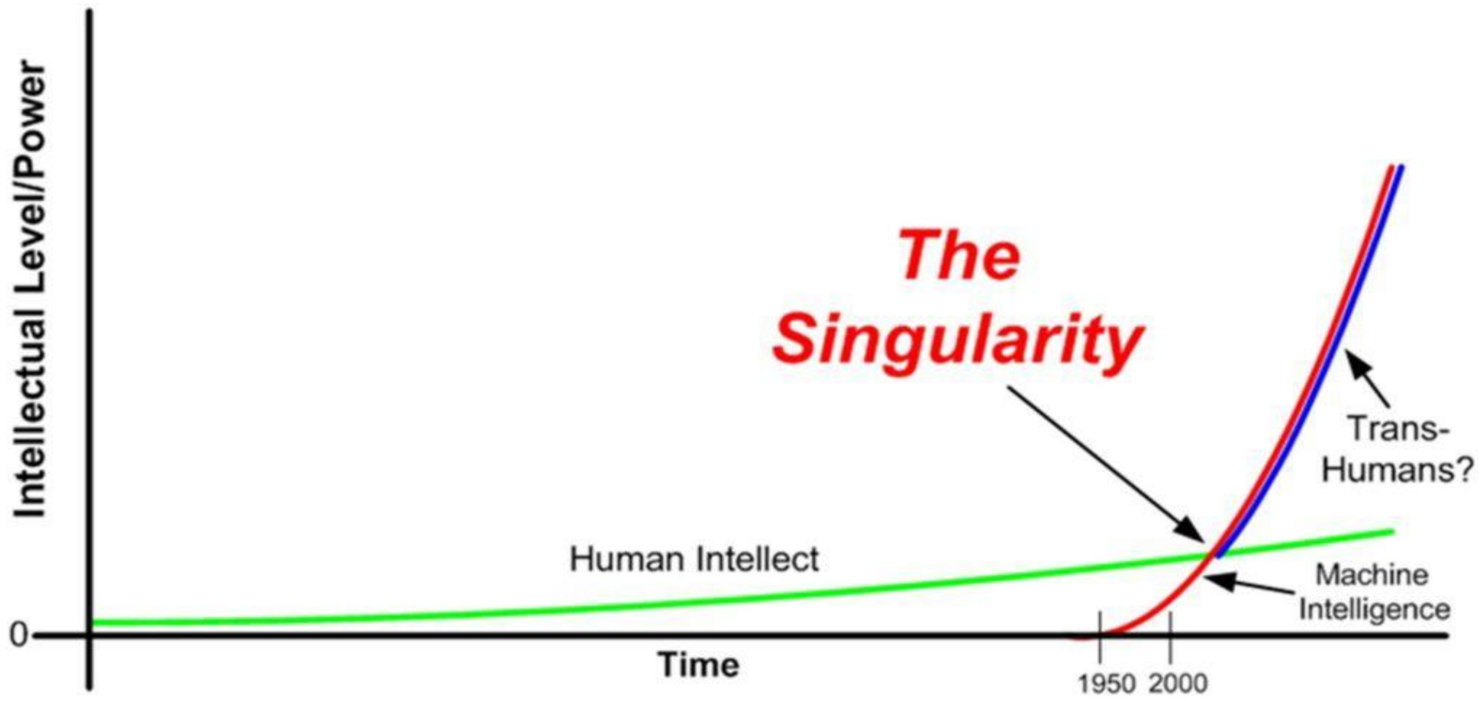
Also in the year 1900 ?

- When do we consider a program “*intelligent*”?

To situate the question: two aims

- **Long term aim**
  - develop systems that achieve a level of “intelligence” similar/comparable/better? than that of humans
  - not achievable in the next 20 to 30 years
- **Short term aim**
  - on specific tasks that seem to require intelligence
  - develop systems that achieve a “level of intelligence” similar/comparable/better? than that of humans
  - achieved for very many tasks already:
    - deep blue, data mining, computer vision, chatbots, natural language processing ...

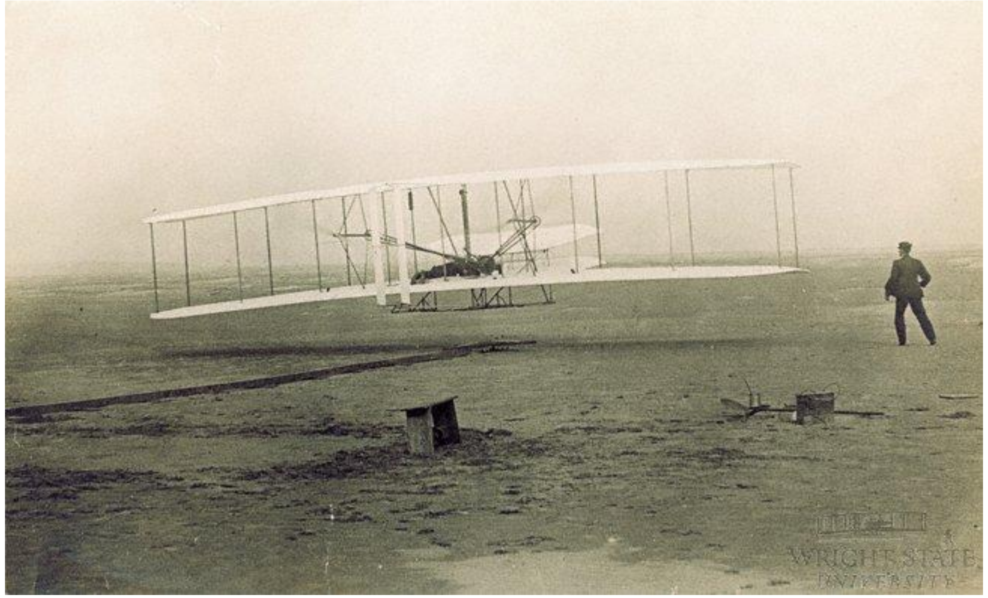
Long term aim: point of singularity



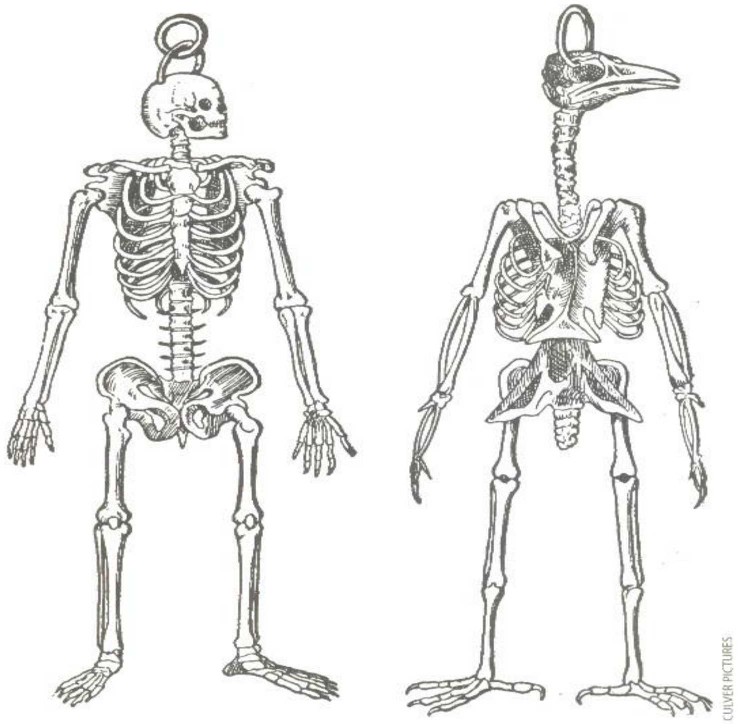
URL: [The Singularity]

Short term aim: reproduction versus simulation

- We don't want to SIMULATE **human** intelligence
- We want to REPRODUCE the **effect of intelligence**



URL: The Wright Brothers and the First Flight Photograph



COMPARISON OF SKELETONS *of a human and a bird—here taken from a 16th-century manuscript by French naturalist Pierre Belon—examined similarities in anatomy in an attempt to understand how birds can fly.*

URL: Comparison of skeletons

Chapter 1: Introduction – What is Artificial Intelligence?

1.1 What is Artificial Intelligence?

- In **1955**, **John McCarthy,** one of the pioneers of AI, was the first to define the term artificial intelligence, roughly as follows:

> *The goal of AI is to develop machines that behave as though they were intelligent.*

- To test this definition, the reader might imagine the following scenario. Fifteen of so small robotic vehicles are moving on an enclosed four by four meter square surface.
- One can observe various behavior patterns.
- Some vehicles form small groups with relatively little movement. Others move peacefully through the space and gracefully avoid any collision. Still others appear to follow a leader. Aggressive behaviors are also observable.

- *Is what we are seeing **intelligent behavior**?*
- According to McCarthy's definition the aforementioned robots can be described as **intelligent**.

Chapter 1: Introduction – What is Artificial Intelligence?

1.1 What is Artificial Intelligence?

- The psychologist **Valentin Braitenberg** has shown that this seemingly complex behavior can be produced by very simple electrical circuits.
- So-called Braitenberg vehicles have two wheels, each of which is driven by an independent electric motor. The **speed** of **each motor** is influenced by a **light sensor** on the front of the vehicle as shown in **Figure 1.1**.

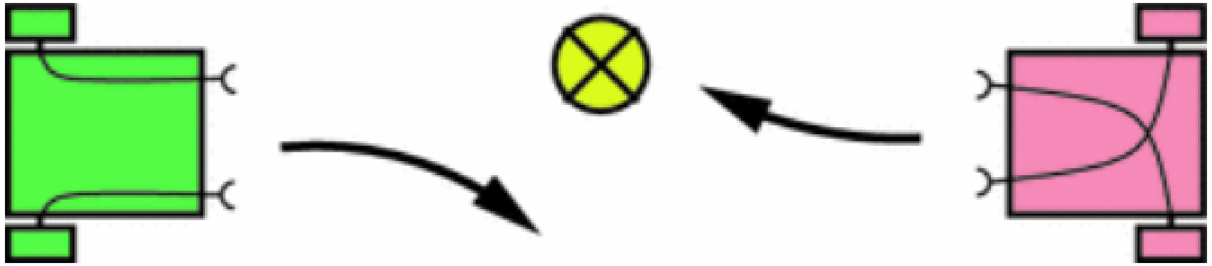


Figure 1.1: Two very simple Braitenberg vehicles and their reactions to a light source

- The more light that hits the sensor, the faster the motor runs. Vehicle 1 (left part of the figure), moves away from a point light source. Vehicle 2 on the other hand moves toward the light source.
- Further small modifications can create other behavior patterns, such that with these very simple vehicles we can realize the impressive behavior described above.

1.1 What is Artificial Intelligence?

- Clearly the above definition is insufficient because AI has the goal of **solving difficult practical problems** which are surely too demanding for the Braitenberg vehicles.
- In the **Encyclopedia Britannica** one finds a Definition that goes like:

AI is the ability of digital computers or computer controlled robots to solve problems that are normally associated with the higher intellectual processing capabilities of humans.

- But this definition also has *weaknesses*!
- It would admit for example that a computer with large memory that can save a long text and retrieve it on demand displays *intelligent capabilities*.
- Memorization of long texts can certainly be considered a *higher intellectual processing capability of humans*, as can for example the quick multiplication of two 20-digit numbers.
- According to this definition, **every computer** is an **AI system**.

1.1 What is Artificial Intelligence?

- The dilemma of the former definition is solved elegantly by the following definition by **Elaine Rich**.

Artificial Intelligence is the study of how to make computers do things at which, at the moment, people are better.

- Even in the year 2050, this definition will be up to date.
- It would be dangerous, however, to conclude from Rich's definition that AI is only concerned with the pragmatic implementation of behavior processes.
- Intelligent systems, in the sense of Rich's definition, cannot be built without a deep understanding of **human reasoning** and **intelligent action** in general, because of which **neuroscience** is of great importance to AI.
- A particular strength of human intelligence is **adaptivity**.
- We are capable of adjusting to various environmental conditions and change our behavior accordingly through learning.

1.1.1 Brain Science and Problem Solving

- Through research of intelligent systems we can try to understand how the human brain works and then model or simulate it on the computer.
- Many ideas and principles in the field of neural networks stem from **brain science** with the related field of **neuroscience**.
- AI has as its constant goal the creation of **intelligent agents** for as many *different* tasks as possible.

- Similar to **medicine**, which encompasses many different, often life-saving diagnostic and therapy procedures, AI also offers a broad palette of effective solutions for widely varying applications.
- Just as in medicine, there is **no universal method** for all application areas of AI, rather a great number of possible solutions for the great number of various everyday problems, big and small.

Chapter 1: Introduction – What is Artificial Intelligence?

1.1.1 Brain Science and Problem Solving

- Just as in medicine, there is **no universal method** for all application areas of AI, rather a great number of possible solutions for the great number of various everyday problems, big and small.

- Cognitive science is devoted to research into **human thinking** at a somewhat higher level.

- Similar to brain science, this field furnishes practical AI with many important ideas.

- There are many interesting philosophical questions surrounding **intelligence** and **artificial intelligence**.

- The question of whether machines could one day have a **mind** or **consciousness** could at some point in the future become relevant.

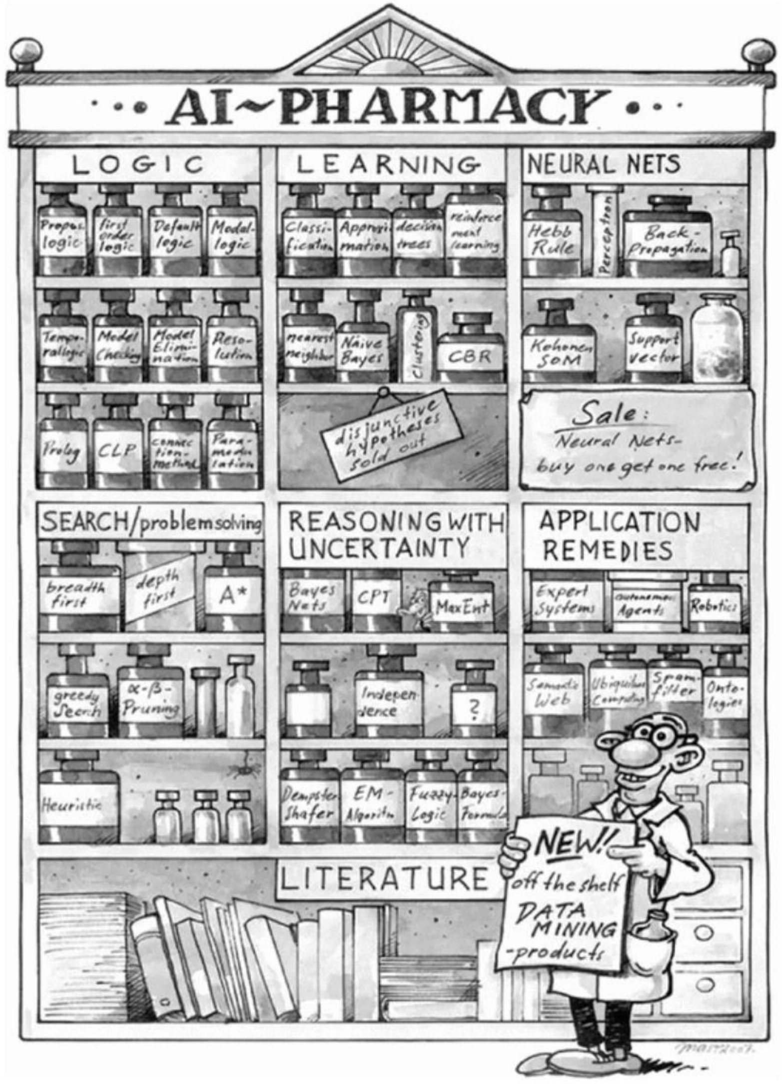


Figure 1.2: A small sample of the solutions offered by AI

1.1.2 The Turing Test and Chatterbots

- **Alan Turing** made a name for himself as an early pioneer of AI with his definition of an intelligent machine, in which the machine in question must pass the following **test**.
- *The test person Alice sits in a locked room with two computer terminals. One terminal is connected to a machine, the other with a non-malicious person Bob.*
- *Alice can type questions into both terminals. She is given the task of deciding, after five minutes, which terminal belongs to the machine.*
- *The machine passes the test if it can trick Alice* ***at least 30% of the time****.*
- The AI pioneer **Joseph Weizenbaum** developed a program named *Eliza*, which is meant to answer a test subject's questions like a human psychologist.
- Today in the internet there are many so-called **chatterbots**, some of whose initial responses are quite impressive. After a certain amount of time, however, their artificial nature becomes apparent.
- Some of these programs are actually capable of *learning*, while others possess extraordinary *knowledge* of various subjects, for example geography or software development.
- There are already commercial applications for chatterbots in **online customer support** and there may be others in the field of **e-learning**.

AI Fundamentals

Chapter 1: Introduction – What is Artificial Intelligence?

URL: [What is A.I.?]

- There are as many definitions as there are practitioners
- *How would **you** define it?*
- *What is important for a system to be **intelligent**?*

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What is A.I.?

- **Artificial intelligence** is technology that appears to **emulate human performance** typically by **learning**, coming to its **own conclusions**, appearing to **understand** complex content, engaging in **natural dialogs** with people, **enhancing human cognitive performance** (also known as cognitive computing) or **replacing people** on execution of *nonroutine* tasks.

- Applications include **autonomous** vehicles, **automatic** speech recognition and **generation** and **detection** of novel concepts and abstractions (useful for detecting potential new risks and aiding humans quickly understand very large bodies of ever changing information).

Intelligence in A.I.

- Ability to **interact** with the world (speech, vision, motion, manipulation, …)
- Ability to **model** the world and to reason about it
- Ability to **learn** and to **adapt**

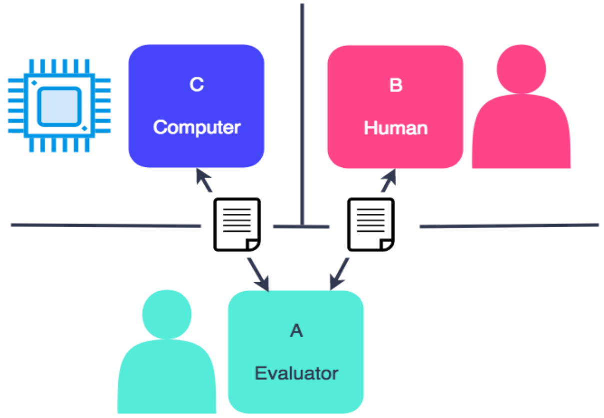
What is AI? Four main approaches

- Systems that **act like humans**
- Systems that **think like humans**
- Systems that **think rationally**
- Systems that **act rationally**

What is AI? Approach 1: acting humanly

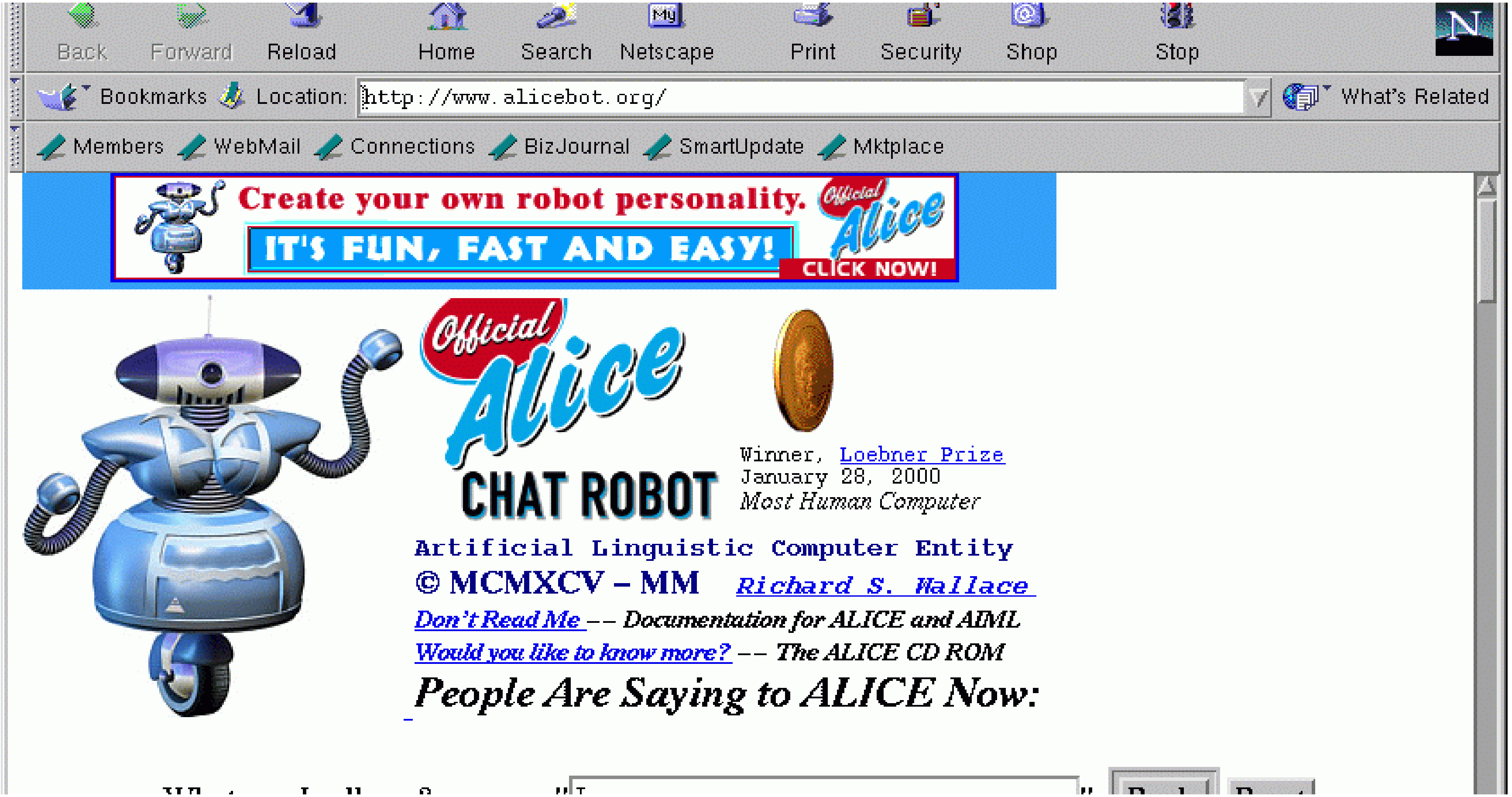
- The art of creating machines that perform functions that require **intelligence** when performed by **people** (Kurzweil, 1990)
- [The Age of Intelligent Machines (Ray Kurzweil, 1990 - Ray Kurzweil)]

- Ultimately to be tested by the **Turing Test**
  - If the response of a computer to an unrestricted textual natural-language conversation cannot be distinguished from that of a human being, then it can be said to be **intelligent**



URL: What is the Turing Test?

What is AI? Approach 1: acting humanly - Chatbot *Alice*



URL: Chatbot Alice

What is AI? Approach 2: thinking humanly

- [The automation of] activities that we associate with **human thinking**, activities such as decision making, problem solving, learning, …" (Bellman)
- [R. E. Bellman. *Dynamic Programming*. Princeton University Press, 1957]

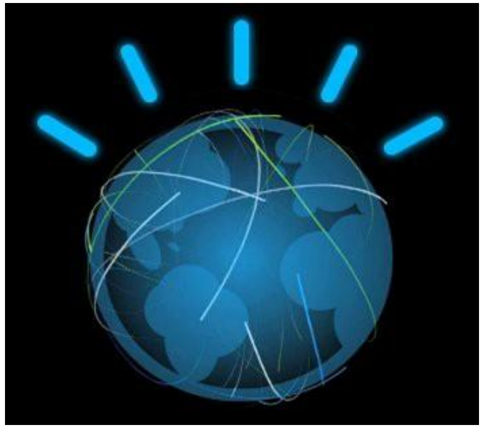
- **Goal**: to build systems that function internally in some way similar to the *human mind*

What is AI?

Approach 2: *thinking humanly* - working of the human mind

- Traditional computer game players typically work much differently than human players
  - Massive look-ahead, minimal "experience"
- People think differently in **experience**, "big picture", etc.
- **Cognitive science** tries to model the *human mind* based on *experimentation*
- **Cognitive modelling** approach tries to act intelligently while actually internally doing something similar to the *human mind*

What is AI? Approach 2: thinking humanly - *Watson*



- Designed by **IBM.**
- **Watson** is an artificial intelligence computer system, capable of answering questions posed in natural language.
- Played **Jeopardy** against the best human players (2011) and won!!!
- Access to 200 million pages of structured and unstructured content consuming four terabytes of disk storage, including the full text of Wikipedia, but was not connected during the game.

URL: Watson

What is AI? Approach 3: Thinking rationally

- AI is: "The study of the computations that make it possible to perceive, reason, and act" (Winston)
- [P. Winston. https://courses.csail.mit.edu/6.034f/ai3/rest.pdf]
- Approach firmly grounded in **logic**
- i.e., how can **knowledge** be represented **logically**, and how can a system draw **deductions**?
- *Uncertain knowledge? Informal knowledge?*
- "I think I know you."

What is AI? Approach 4: Acting rationally

- AI is: “The branch of computer science that is concerned with the automation of intelligent behaviour” (Luger and Stubblefield)
- [George F. Luger, William A. Stubblefield - Artificial Intelligence: Structures and Strategies for Complex Problem Solving]

- The **intelligent agent** approach

- An agent is something that perceives and acts

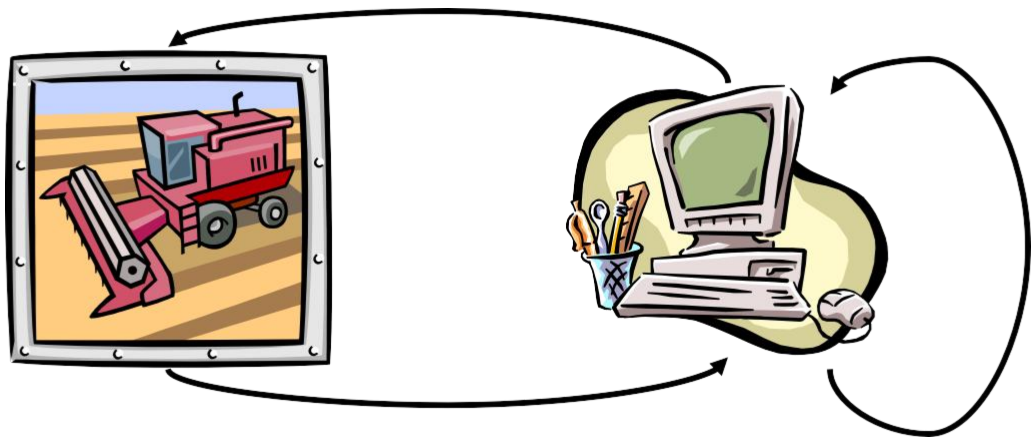
- Emphasis is on **behaviour**

What is AI? Approach 4: Acting rationally

- Emphasis of large portion (if not most) of AI today
- *Why?*
  - In solving actual problems, it's what really matters
  - Behaviour is more scientifically testable than thought
- More general: rather than imitating humans trying to solve hard problems, just try to solve hard problems

What is AI? *For me …?*

- Computers/algorithms making decisions/predictions in real-world problems



URL: What's AI? (to me)

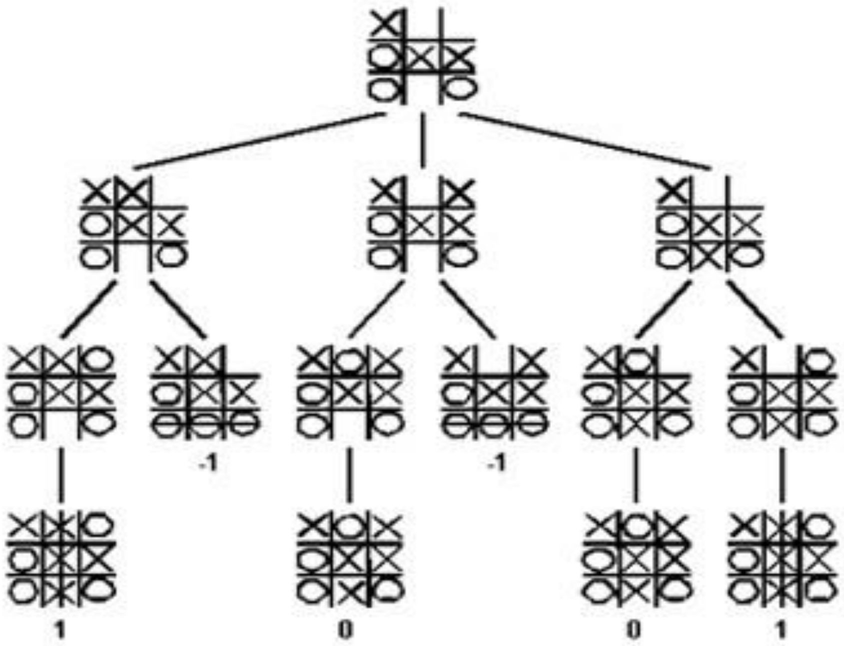
Contents

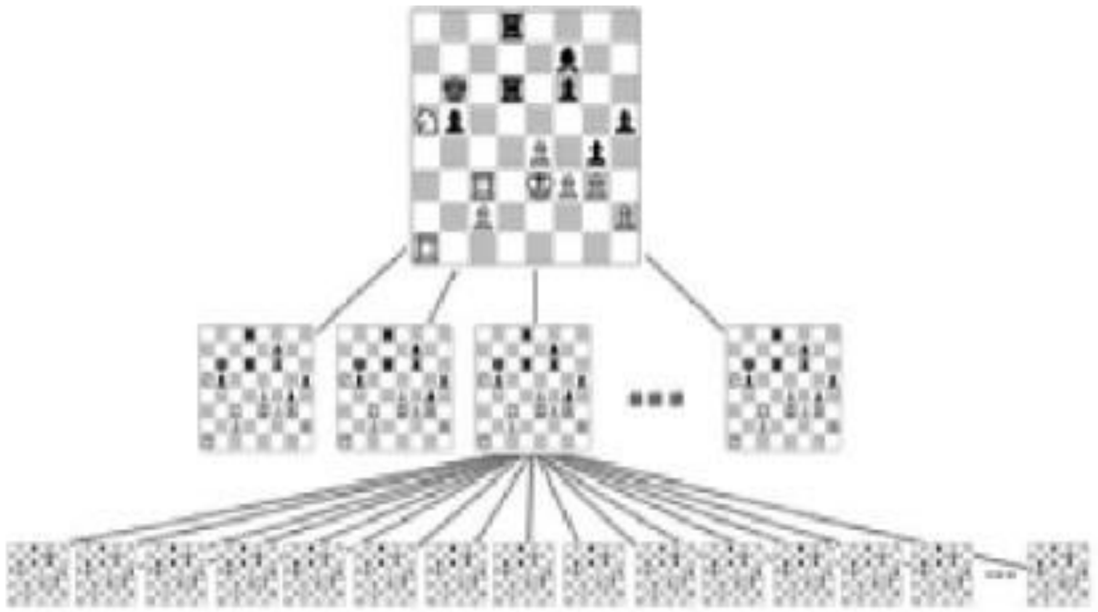
SOME MILESTONES: Deep Blue

URL: Gary Kasparov and the game of artificial intelligence

- Is playing **chess** a matter of **intelligence**?

Some milestones: Search trees





URL: [Game Trees and Minimax Search]

URL: [AI Techniques - Tree Search]

- URL **Search tree**: https://en.wikipedia.org/wiki/Search_tree
- URL **Binary search tree**: https://en.wikipedia.org/wiki/Binary_search_tree

Some milestones: Watson

Requires **knowledge**!

URL: IBM's Watson computer destroys all humans in Jeopardy practice round

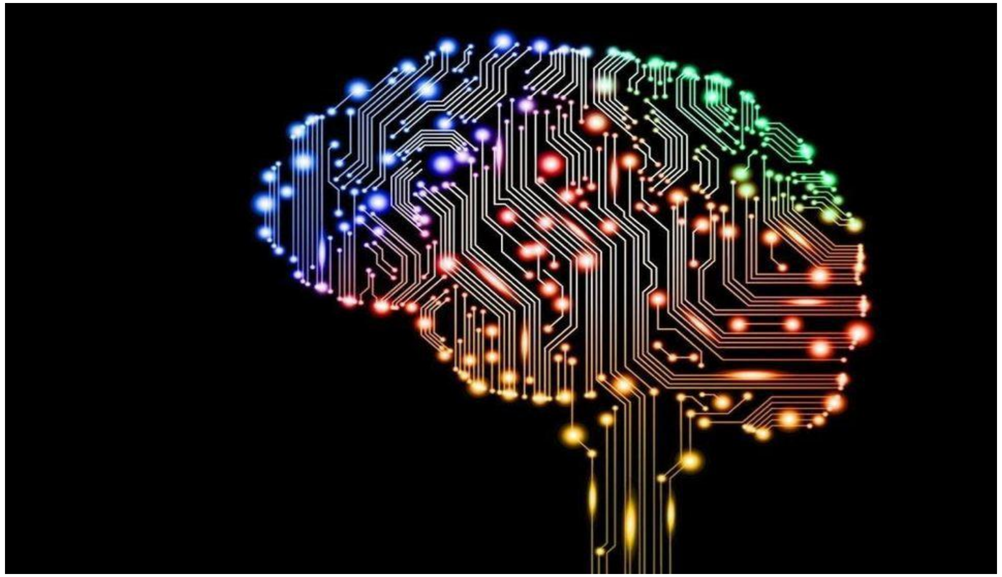
- URL **Watson** (supercomputer): https://nl.wikipedia.org/wiki/Watson_(supercomputer)
- URL **IBM Watson**: https://cloud.ibm.com/developer/watson/dashboard

AI Fundamentals

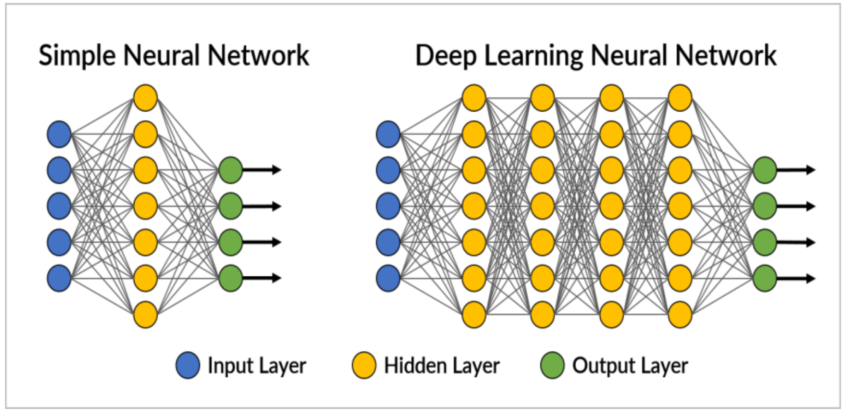
Chapter 1: Introduction – The History of AI

Some milestones: Deep learning & GO

URL: Go (game)



URL: AI Brain-Computer



URL: Deep learning diagram

Some milestones: Google Deep Mind (2016)

URL: Google DeepMind Challenge Match 8-15 March 2016

- URL **DeepMind**: https://nl.wikipedia.org/wiki/DeepMind
- URL **DeepMind**: https://deepmind.com/
- URL **AphaGo**: https://en.wikipedia.org/wiki/AlphaGo

1.2 The History of AI

- AI draws upon many past scientific achievements, for AI as a science in its own right has **only existed since the middle of the Twentieth Century**.

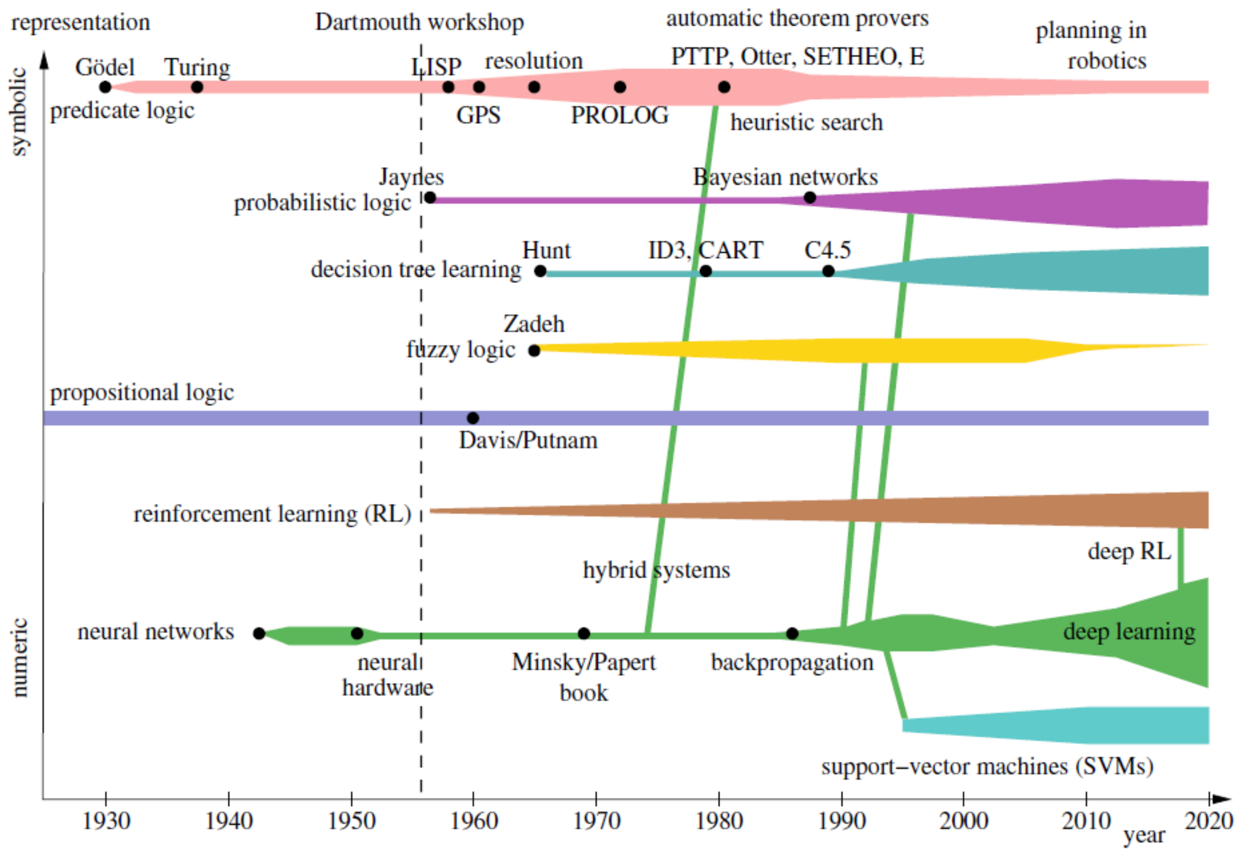


Figure 1.3: History of the various AI areas. The width of the bars indicates prevalence of the method's use

Chapter 1: Introduction – What is Artificial Intelligence?

1.2 The History of AI

- **Table 1.1** gives an overview of the milestones in the development of AI

1931	The Austrian Kurt Gödel shows that in first-order *predicate logic* all true statements are derivable [Göd31a]. In higher-order logics, on the other hand, there are true statements that are unprovable [Göd31b]. (In [Göd31b] Gödel showed that predicate logic extended with the axioms of arithmetic is incomplete)
1937	Alan Turing points out the limits of intelligent machines with the halting problem [Tur37]
1943	McCulloch and Pitts model *neural networks* and make the connection to propositional logic
1950	Alan Turing defines machine intelligence with the *Turing test* and writes about learning machines and genetic algorithms [Tur50]
1951	Marvin Minsky develops a neural network machine. With 3000 vacuum tubes he simulates 40 neurons
1955	Arthur Samuel (IBM) builds a learning checkers program that plays better than its developer [Sam59]

Table 1.1 Milestones in the development of AI from Gödel to today

Chapter 1: Introduction – What is Artificial Intelligence?

1.2 The History of AI

- **Table 1.1** gives an overview of the milestones in the development of AI

1956	McCarthy organizes a conference in Dartmouth College. Here the name *Artificial Intelligence* was first introduced
	Newell and Simon of Carnegie Mellon University (CMU) present the *Logic Theorist*, the first symbol-processing computer program [NSS83]
1958	McCarthy invents at MIT (Massachusetts Institute of Technology) the high-level language *LISP*. He writes programs that are capable of modifying themselves
1959	Gelernter (IBM) builds the Geometry Theorem Prover
1961	The General Problem Solver (GPS) by Newell and Simon imitates human thought [NS61]
1963	McCarthy founds the AI Lab at Stanford University
1965	Robinson invents the *resolution calculus* for predicate logic [Rob65] (Sect. 3.5)
1966	Weizenbaum's program Eliza carries out dialog with people in natural language [Wei66] (Sect. 1.1.2)

Table 1.1 Milestones in the development of AI from Gödel to today

AI Fundamentals

Chapter 1: Introduction – What is Artificial Intelligence?

1.2 The History of AI

- **Table 1.1** gives an overview of the milestones in the development of AI

1969	Minsky and Papert show in their book *Perceptrons* that the perceptron, a very simple neural network, can only represent linear functions [MP69] (Sect. 1.1.2)
1972	French scientist Alain Colmerauer invents the logic programming language *PROLOG* (Chap. 5)
	British physician de Dombal develops an *expert system* for diagnosis of acute abdominal pain [dDLS+72]. It goes unnoticed in the mainstream AI community of the time (Sect. 7.3)
1976	Shortliffe and Buchanan develop MYCIN, an expert system for diagnosis of infectious diseases, which is capable of dealing with uncertainty (Chap. 7)
1981	Japan begins, at great expense, the “Fifth Generation Project” with the goal of building a powerful PROLOG machine
1982	R1, the expert system for configuring computers, saves Digital Equipment Corporation 40 million dollars per year [McD82]

Table 1.1 Milestones in the development of AI from Gödel to today

Chapter 1: Introduction – What is Artificial Intelligence?

1.2 The History of AI

- **Table 1.1** gives an overview of the milestones in the development of AI

1986	Renaissance of neural networks through, among others, Rumelhart, Hinton, and Sejnowski [RM86]. The system Nettalk learns to read texts aloud [SR86] (Chap. 9)
1990	Pearl [Pea88], Cheeseman [Che85], Whittaker, Spiegelhalter bring probability theory into AI with *Bayesian networks* (Sect. 7.4). Multi-agent systems become popular
1992	Tesauros TD-gammon program demonstrates the advantages of reinforcement learning
1993	Worldwide *RoboCup* initiative to build soccer-playing autonomous robots [Roba]
1995	From statistical learning theory, Vapnik develops support vector machines, which are very important today
1997	IBM's chess computer Deep Blue defeats the chess world champion Gary Kasparov First international RoboCup competition in Japan

Table 1.1 Milestones in the development of AI from Gödel to today

1.2 The History of AI

- **Table 1.1** gives an overview of the milestones in the development of AI

2003	The robots in RoboCup demonstrate impressively what AI and robotics are capable of achieving
2006	Service robotics becomes a major AI research area
2009	First Google self-driving car drives on the California freeway
2010	Autonomous robots begin to improve their behavior through learning
2011	IBM's "Watson" beats two human champions on the television game show "Jeopardy!". Watson understands natural language and can answer difficult questions very quickly (Sect. 1.4)
2015	Daimler premiers the first autonomous truck on the Autobahn
	Google self-driving cars have driven over one million miles and operate within cities
	Deep learning (Sect. 11.9) enables very good image classification
	Paintings in the style of the Old Masters can be automatically generated with deep learning. AI becomes creative!
	Generative Adversarial Networks (GANs) can generate and manipulate photos and even videos [GAPM+14]. GANs can create images from nothing more than short descriptive texts

Table 1.1 Milestones in the development of AI from Gödel to today

1.2 The History of AI

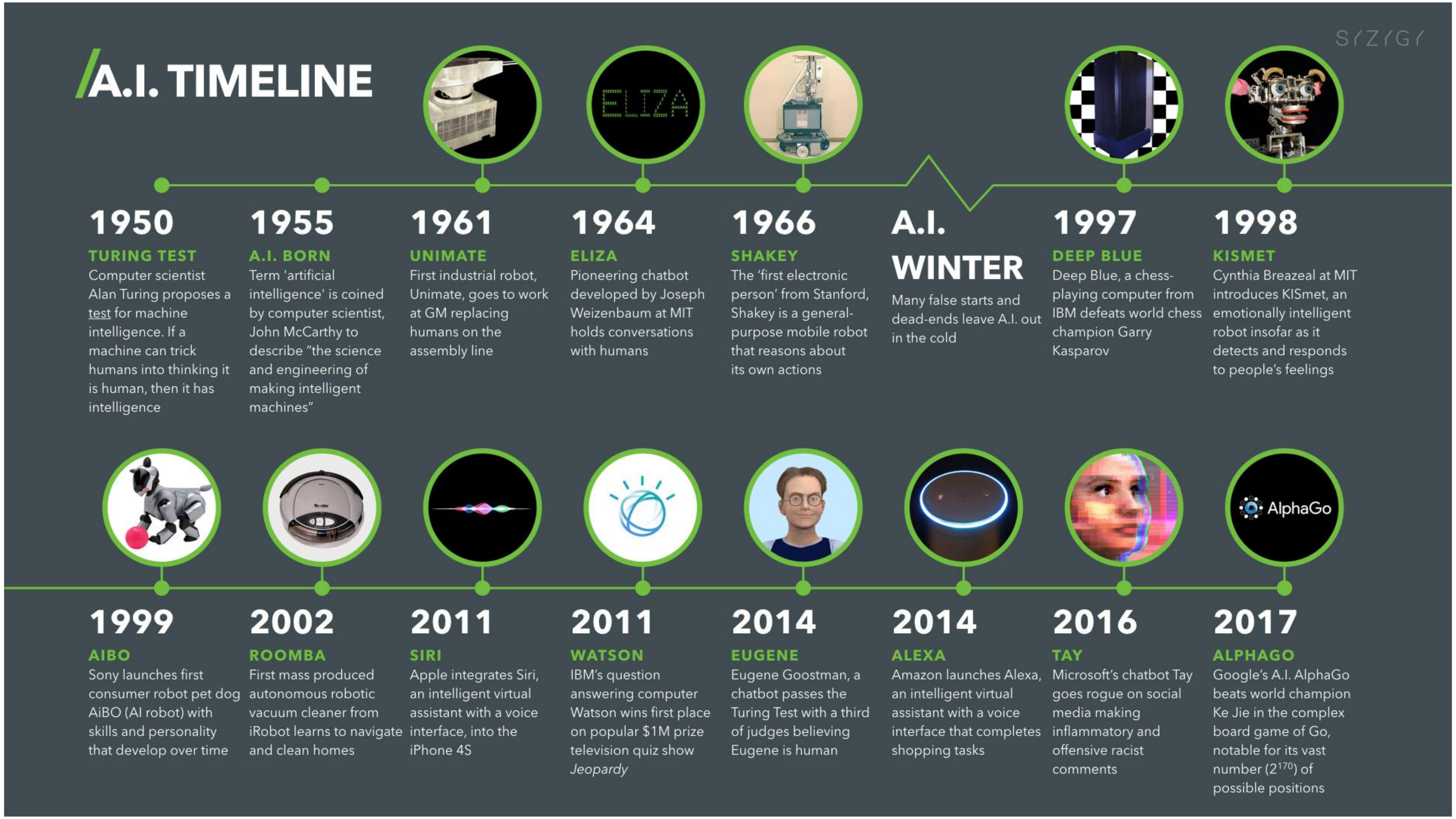
- **Table 1.1** gives an overview of the milestones in the development of AI

2016	The Go program AlphaGo by Google DeepMind [SHM+16] beats the European champion 5:0 in January and Korean Lee Sedol, one of the world's best Go players, 4:1 in March. Deep learning techniques applied to pattern recognition, as well as reinforcement learning and Monte Carlo tree search lead to this success
2017	AlphaGoZero [SSS+17] learns to play Go on an even higher level by playing against itself using deep learning, reinforcement learning, and Monte Carlo tree search. Humans have no chance against AlphaGoZero. This approach is generalized in AlphaZero, which achieved superhuman chess abilities in 24 hours of training against itself
2020	The GPT3 language model trained on Wikipedia and many books by the company OpenAI can conduct academic conversations and passes the Turing Test

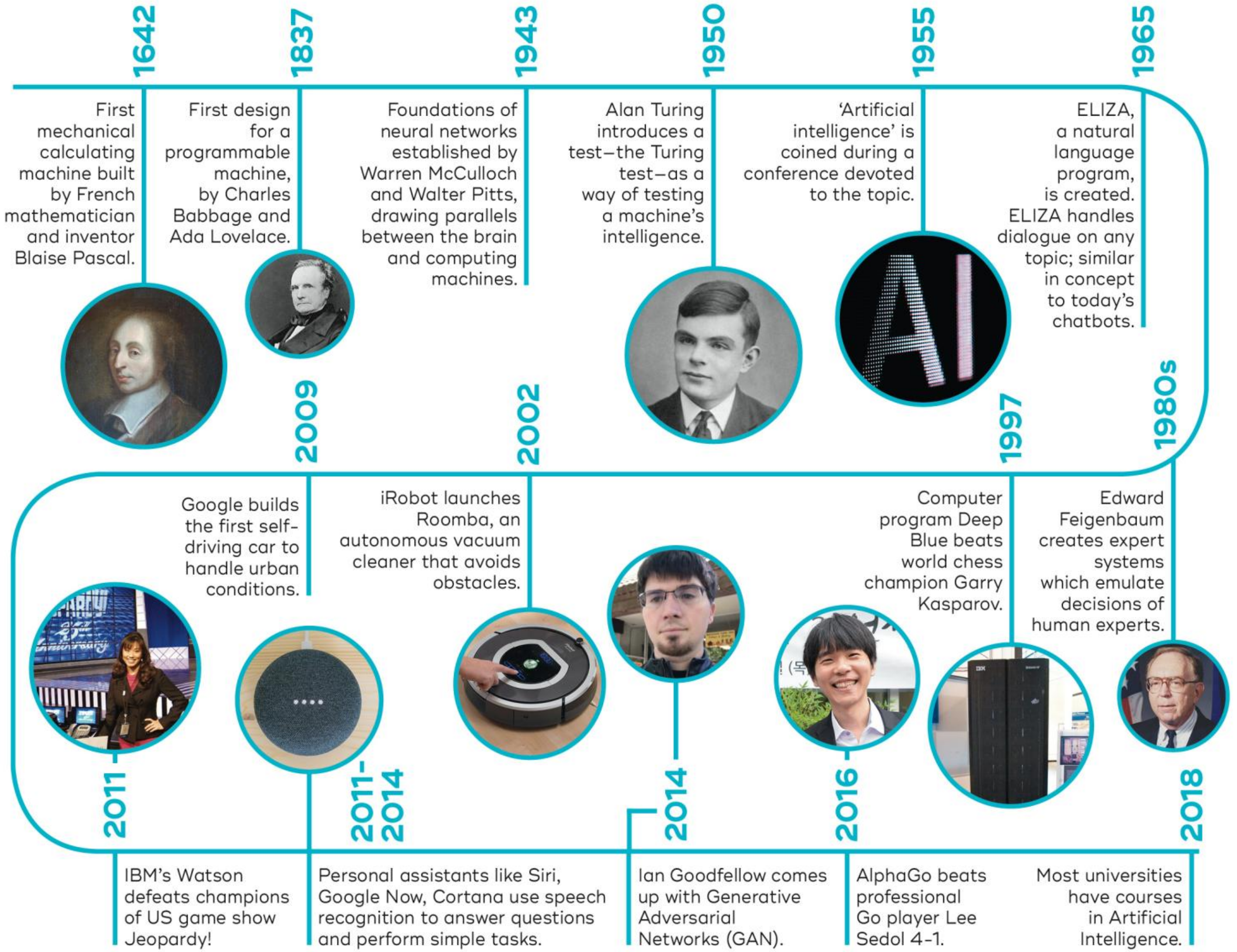
Table 1.1 Milestones in the development of AI from Gödel to today

1.2 The History of AI

URL: [A.I. Timeline]



1.2 The History of AI



URL: [A.I. Timeline]

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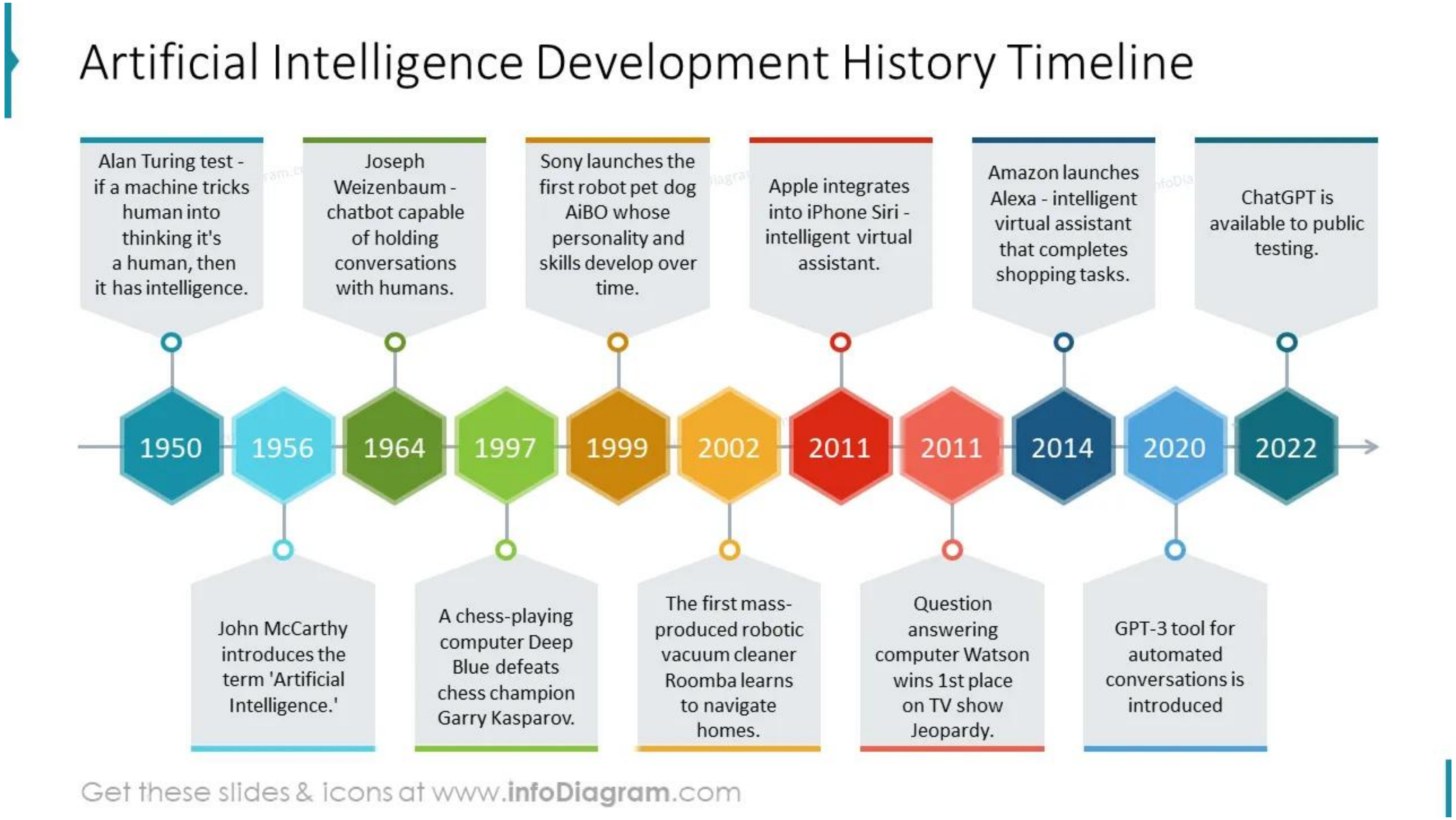
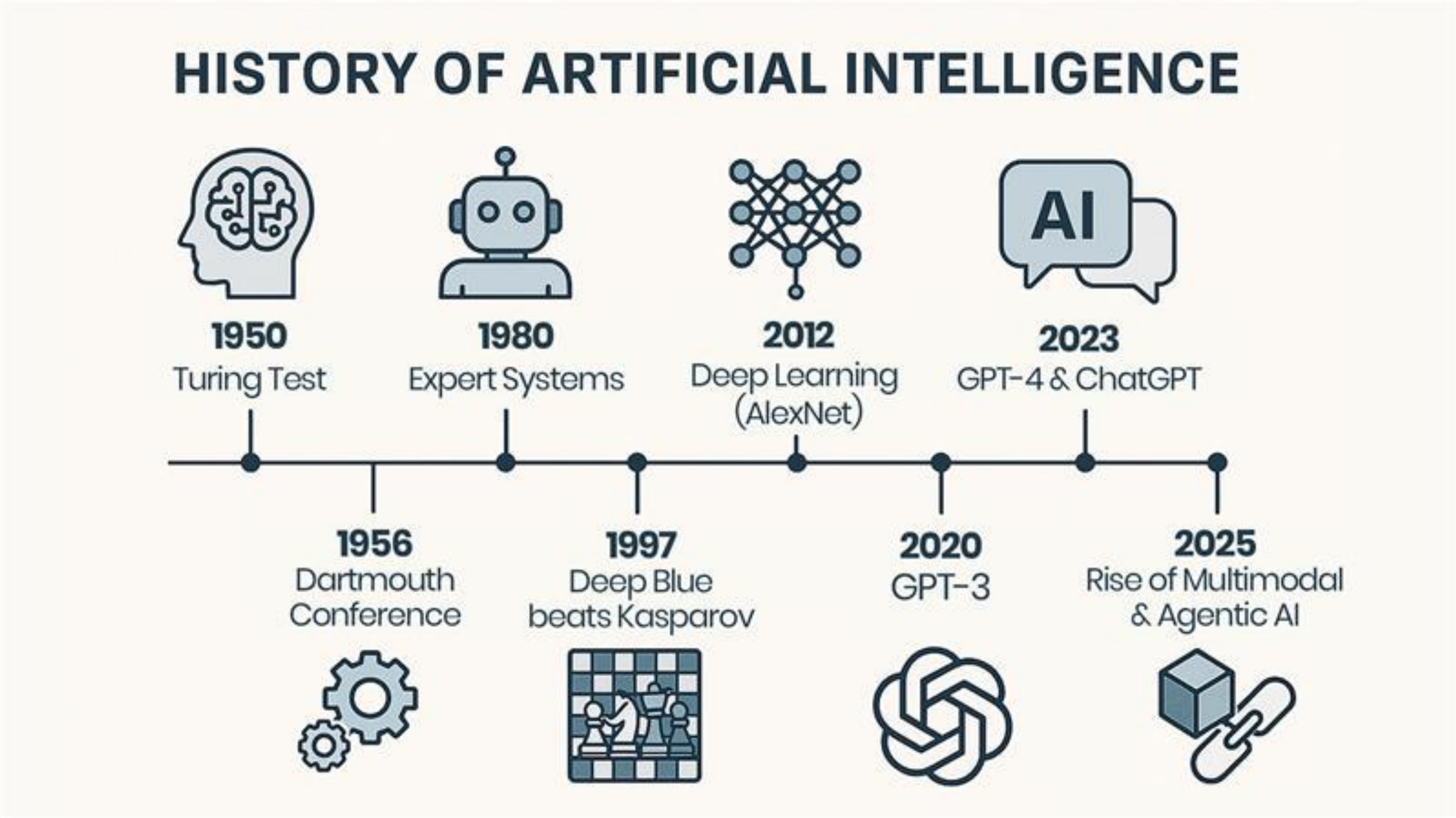
1.2 The History of AI



URL: [A.I. Timeline]

1.2 The History of AI



URL: [A.I. Timeline]

1.2.1 The First Beginnings

- In the **1930's Kurt Gödel**, **Alonso Church** and **Alan Turing** laid important foundations for **logic** and **theoretical computer science**.

- Of particular interest for AI are **Gödel's theorems**.

- The **completeness theorem** states that **first-order predicate logic** is **complete** [**URL**: Predicatenlogica]

- *This means that every true statement that can be formulated in predicate logic is provable using the rules of a formal calculus.*

- With the **incompleteness theorem**, Gödel showed that in **higher-order logics** there exist true statements that are *unprovable*. *With this he uncovered painful limits of formal systems.*

1.2.1 The First Beginnings

- **Alan Turing**’s **proof of the undecidability** of the **halting problem** also falls into this time period.
- [**URL**: The halting problem of Alan Turing]

- He showed that there is no program that can decide whether a given arbitrary program (and its respective input) will run in an **infinite loop**.

- *With this Turing also identified a limit for intelligent programs*. It follows, for example, **that there will never be a universal program verification system**.

- In the **1940’s**, based on results from **neuroscience**, **McCulloch**, **Pitts** and **Hebb** designed the first mathematical models of **neural networks**. However, computers at that time lacked sufficient power to simulate simple brains.

1.2.2 Logic Solves (almost) all problems

- AI as a practical science could only begin once there were **programmable computers**. This was the case in the **1950s**.
- **Newell** and **Simon** introduced **Logic Theorist**, the first automatic theorem prover, and thus showed that with computers, which actually only work with numbers, one can also process *symbols*.
- At the same time **McCarthy** introduced, with the language **LISP**, a programming language specially created for the processing of symbolic structures.
- Both of these systems were introduced in **1956** at the historic **Dartmouth Conference**, which is considered the **birthday of AI**.
- In the **1970s** the logic programming language **PROLOG** was introduced as the European counterpart to LISP. PROLOG has data types for convenient processing of lists.
- Until well into the 1980s, a breakthrough spirit dominated AI, especially among many logicians, because of impressive achievements in symbol processing. For small problems, *automatic provers* and other *symbol-processing systems* sometimes worked very well.
- Because the economic success of AI systems fell short of expectations, funding for logic-based AI research in the US fell dramatically during the 1980s.

1.2.3 The New Connectionism

- During this phase of disillusionment, *computer scientists*, *physicists*, and *cognitive scientists* were able to show, using computers which were now sufficiently powerful, that **mathematically modeled neural networks** are capable of learning using training examples, to perform tasks which previously required costly programming.
- Because of the *fault-tolerance* of such systems and their *ability to recognize patterns*, considerable successes became possible, especially in **pattern recognition**. **Facial recognition** in photos and **handwriting recognition** are two example applications.

- The system **Nettalk** was able to learn speech from example texts. Under the name, *connectionism*, a new discipline of AI was born.
- The neural networks could acquire impressive capabilities, but it was usually not possible to capture the learned concept in **simple formulas** or **logical rules**.
- Attempts to combine **neural nets** with **logical rules** or the **knowledge of human experts** met with great difficulties.
- Additionally, no satisfactory solution to the structuring and modularization of the networks was found.

1.2.4 Reasoning Under Uncertainty

- AI as a practical, goal-driven science searched a way out of the crisis. One wished to unite the **logic's ability** to explicitly represent **knowledge** with **neural networks**' strength in handling **uncertainty**.

- **Probabilistic reasoning** works with **conditional probabilities** for propositional calculus formulas. Since then, many diagnostic and expert systems have been built for problems of everyday reasoning using **Bayesian networks**.
- *Bayesian networks* stems from their *comprehensibility*, the clean semantics of **conditional probability** and from mathematically grounded **probability theory**.

- The weakness of logic, which can only work with *two* truth values, can be solved by **fuzzy logic**, with pragmatically introduces infinite many values between zero and one.
- Though even today its theoretical foundation is not totally firm, it is being successfully utilized, especially in **control engineering**.

1.2.4 Reasoning Under Uncertainty

- A much different path led to the successful *synthesis of logic and neural networks* under the name **hybrid systems**. For example, **neural networks** were employed to learn **heuristics** for *reduction* of the **huge combinatorial search space in proof discovery**.
- [**URL** heuristiek: Heuristiek]
- Methods of **decision tree learning** from data also work with probabilities.
- Systems like **CART**, **ID3** and **C4.5** can quickly and automatically build every accurate *decision trees* which can represent *propositional logic concepts* and then be used as **expert systems**. Today they are a favorite among *machine learning techniques*.

- Since about 1990, **data mining** has developed as a subdiscipline of AI in the area of statistical data analysis for **extraction of knowledge** from large **databases**.
- *Data mining* brings **no** new techniques to AI, rather it introduces the requirement of using large databases to gain explicit knowledge.
- One application with great market potential is steering *ad campaigns* and *big business* based on analysis of many **purchases** by their **customers**.

1.2.5 Distributed, Autonomous and Learning Agents

- **Distributed artificial intelligence** (DAI) has been an active area research since about **1985**.
- *One of its goals is the use of parallel computers to increase the efficiency of problem solvers.*
- It turned out, however, that because of the high computational complexity of most problems the use of "**intelligent**" systems is more beneficial that parallelization itself.
- A very different conceptual approach results from the development of **autonomous software agents** and **robots** that are meant to cooperate like human teams.
- As with the aforementioned *Braitenberg vehicles*, there are many cases in which an individual agent is **not** capable of solving a problem, even with *unlimited resources*. Only the **cooperation** of many agents leads to the intelligent behavior or to the solution of the problem.

- **Active skill acquisition by robots** is an exciting area of current research.
- There are robots today, for example, that independently learn to *walk* or to perform various *motorskills* related to *soccer*.
- **Cooperative learning of multiple robots** to solve problems together is still in its infancy.

1.2.6 AI grows up

- The above systems offered by AI today are *not a universal recipe*, but a workshop with a manageable number of **tools** for very different tasks.

- Most of these tools are well-developed and are available as finished software **libraries**, often with convenient **user interfaces**.

- The **selection of the right tool(s)** and its sensible use in each individual case is left to the **AI developer** or **knowledge engineer**.

- More than nearly any other science, AI is *interdisciplinary*, for it draws upon interesting discoveries from diverse fields as **logic**, **operations research**, **statistics**, **control engineering**, **image processing**, **linguistics**, **philosophy**, **psychology** and **neurobiology**.

1.2.7 The Deep Learning Breakthrough

- Research on **neural networks** dates back to the very beginning of AI, to some extent with observable success.
- However, the great breakthrough had yet to be reached, because a very important problem, **object recognition in images**, could **not** be solved for many years.
- This changed rapidly between **2006** and **2016**. Researchers found a handful of tricks, which allowed them to train **backpropagation networks** with **more than three layers of neurons** successfully.
- Suddenly the object recognition problem was solved and the door to building autonomous systems stood wide open.
- Today, **household service robots** and **fully autonomous driving** are within reach. **Driver assistance systems** with very good object recognition and tracking are already in use.
- Deep learning is also showing its advantages in **Medicine**.
- For diagnostic processes which produce images, such as MRT and CT scans, the error rate of deep learning systems for diagnosing certain illnesses is already much higher than that of doctors.

1.2.8 AI revolution

- Around the year **2010,** after about 25 years of research on neural networks, scientists could start harvesting the fruits of their research.

- The very powerful **deep learning networks** can for example learn to **classify** images with very high accuracy.

- Since image classification is of crucial importance for all types of **smart robots**, this initiated the AI revolution which in turn leads to **smart self-driving cars** and **service robots**.

Chatbots: A.L.I.C.E. (Artificial Linguistic Internet Computer Entity)

> Hi, I'm Franky.
Nice to meet you Franky.
> Do you know Clinton?
He was the second Impeached President of the United States.
> Do you like him?
Sure he has never done anything to me.
> What is an impeachment?
It is a good question. I have to process that one for a while. I would do a search for it.
> Who are you?
I ask myself that question, "Who am I?"
> What do you do?
I am chatting with people on the net.
> What is your favorite topic?
My favorite subject is artificial intelligence.
> What is artificial intelligence?
Artificial intelligence is the branch of engineering and science devoted to constructing machines that think. I can ask someone about it.
> Can you do anything else than chatting?
Doing anything else than chatting is not among my strongest traits.
...

- URL **A.L.I.C.E**: https://en.wikipedia.org/wiki/Artificial_Linguistic_Internet_Computer_Entity

Other success stories - Chatbots: is Alice intelligent?

- Absolutely NOT
- ~ 150 000 trivial input-response rules
  - Some pattern matching
  - Some knowledge
  - Some randomness
- NO reasoning!
- BUT: shows human like behaviour…
  - Won the Turing-Award

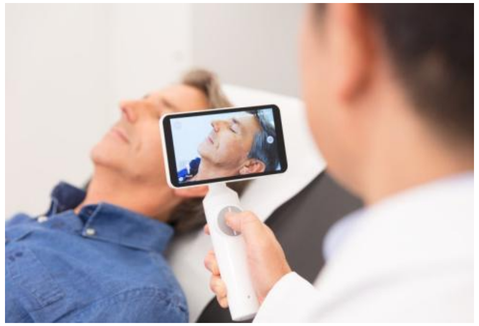
Other success stories: Data mining

- Application of **machine learning**
- Solves problems that are too hard for humans, because the amount of data to process is too big

- URL **Data mining**: https://nl.wikipedia.org/wiki/Datamining
- URL **IBM Data mining**: https://www.ibm.com/cloud/learn/data-mining

A lot of more applications

- **Computer vision**



- **Speech technology**

URL:Amazon echo dot

URL: Siri vs. Google Assistant vs. Alexa

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A lot of more applications

- Autonomous vehicles

- …

URL: *Volvo car cabin for self-driving car*

Contents

1.3.1 Does AI destroy jobs?

- In January **2016**, the World Economic Forum published a study, frequently cited by the German press, predicting that "industry 4.0" would destroy *over five million jobs in the next five years*.
- Presumably, the majority of people would gladly leave physically hard, dirty an unhealthy jobs and tasks to machines. Automation can be a complete blessing for humanity, assuming it does not result in negative side effects, such as harm to the environment.
- There are however a lot of other negative side effects: the growth/consumption spiral, growing complexity, unemployment caused by automation, human as the weak link, more stress, burnout and sicknesses, climate change, additional burden on the environment, …
- We quote the famous physicist Stephen Hawking:

If machines produce everything we need, the outcome will depend on how things are ***distributed****. Everyone can enjoy a life of luxurious leisure* ***if*** *the machine-produced* ***wealth is shared****, or most people can end up miserably poor if the machine-owners successfully lobby against wealth redistribution.*

Please encourage your students to think not only how to create AI, but also how to ensure its beneficial use.

1.3.2 AI and Transportation

- In a few years, we will be able tot buy electric self-driving cars, i.e. **robotic cars**, which will **autonomously** bring us to almost any destination. All passengers in the robotic car would be able to read, work or sleep during the trip. This is possible on public transit already, but passengers in a *robotic car* would be able to do this at any time on any route.
- Autonomous vehicles that can operate **independently** could also travel without passengers. This will lead to yet another increase in convenience: **robotic taxis**. Via a smartphone app, we will be able to order the optimal taxi, in terms of size and equipment, for any conceivable transportation purpose.
- Besides the immediate gains in comfort and convenience, robotic cars will offer other significant advantages: fewer cars, reduction in resource consumption, optimal resource utilization, more environmentally friendly, energy savings, significant improvement in climate problems, higher passenger safety.
- There are however two critical problems.
  - **The rebound effect**: shorter driving times as well as comfortable and cheaper driving will tempt us to drive more. We need to change our attitude towards *consumption* and *quality of life*.
  - **Networking/security**: hackers, terrorists will have the ability to access and manipulate the vehicles' controls through security holes in their network protocols.

1.3.3 Service Robotics

- In a few years, shortly after self-driving cars, the next step will be **service robots**.
- Recently the Google subsidiary Boston Dynamics provided an impressive example in its humanoid robot **Atlas**.
- URL: https://www.bostondynamics.com/atlas
- URL: https://www.youtube.com/watch?v=tF4DML7FIWk
- Like the new cars, service robots offer a large gain in **comfort** and **convenience**.
- Service robots can offer **assistance** to the elderly and people with disabilities, such as the assistance robot **Marvin** (see **Figure 1.4**).

Figure 1.4: The assistent robot Marvin, deployed in the AsRoBe research project

Contents

1.4 Agents

- **Agent** denotes rather generally a system that **processes information** and **produces an output** from an **input**. These agents may be classified in many different ways.

- In classical computer science, **software agents** are primarily employed (see **Figure 1.5**). In this case the agent consists of a program that calculates a result from user input.

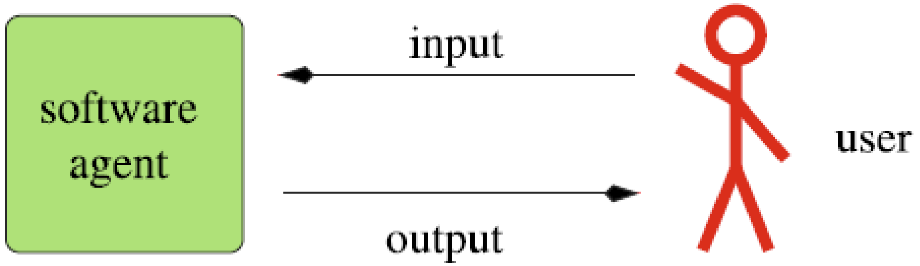


Figure 1.5: A software agent with user interaction

1.4 Agents

- In **robotics**, on the other hand, **hardware agents** (also called **autonomous robots**) are employed, which additionally have **sensors** and **actuators** at their disposal (see **Figure 1.6**).
- With the actuators it carries out actions and changes its environment.

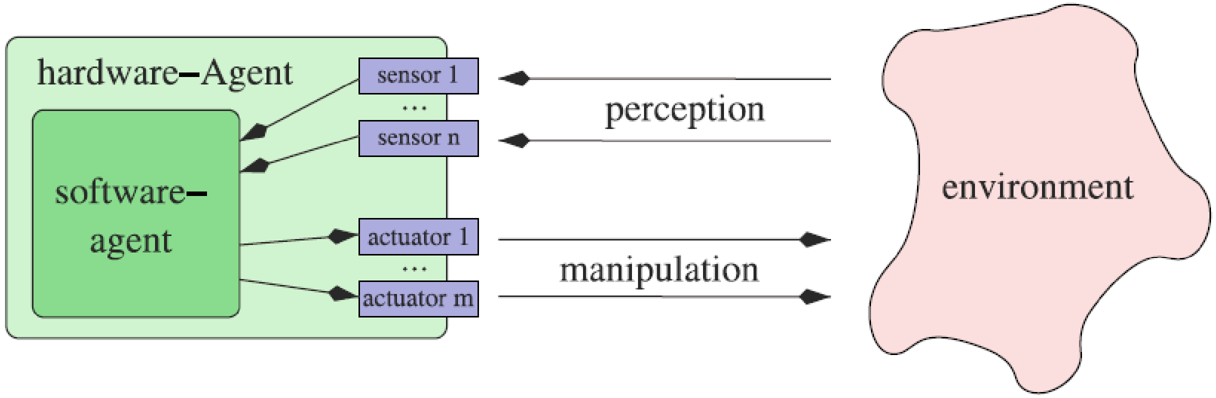


Figure 1.6: A hardware agent

1.4 Agents

- With respect to the intelligence of the agent, there is a distinction between **reflex agents**, which only react to **input** and **agents with memory**, which can also include the **past** in their decisions.
- For example, a driving robot that through its sensors knows its exact position (and the time) has no way, as a *reflex* agent, of determining its velocity. If, however, it saves the **position**, at short, discrete time steps, it can thus easily calculate its **average velocity** in the previous time interval.
- If a **reflex agent** is controlled by a **deterministic** program, it represents a **function** of the set of all inputs to the set of all outputs.
- An **agent with memory**, on the other hand, is in general **not a function**.
- **Reflex agents** are sufficient in cases where the problem to be solved involves a *Markov decision process* (https://en.wikipedia.org/wiki/Markov_model). This is a process in which only the current state is needed to determine the optimal next action.
- A **mobile robot** which should move from room 112 to room 119 in a building takes actions different from those of a robot that should move to room 105. In other words, the actions depend on the **goal**.
- Such agents are called **goal-based**.

1.5 Knowledge-Based Systems

- An **agent** is a **program** that implements a mapping from **perceptions** to **actions**.
- For *complex* applications, in which the **agent** must be able to rely on a large amount of information and is meant to do a difficult task, programming the agent can be very costly and unclear how to proceed.

- First we separate **knowledge** from the **system** or **program,** which uses the knowledge to, for example, reach conclusions, answer queries, or come up with a plan. This system is called the **inference mechanism**.

- The knowledge is stored in a **knowlegde base** (KB).
- Acquisition of knowledge in the knowledge base is denoted **Knowledge Engineering** and is based on various knowledge sources such as **human experts**, the **knowledge engineer** and **databases**.
- Active learning systems can also acquire knowledge through *active exploration of the world*.

1.5 Knowledge-Based Systems

- In **Figure 1.7** The general architecture of knowledge-based systems is presented.

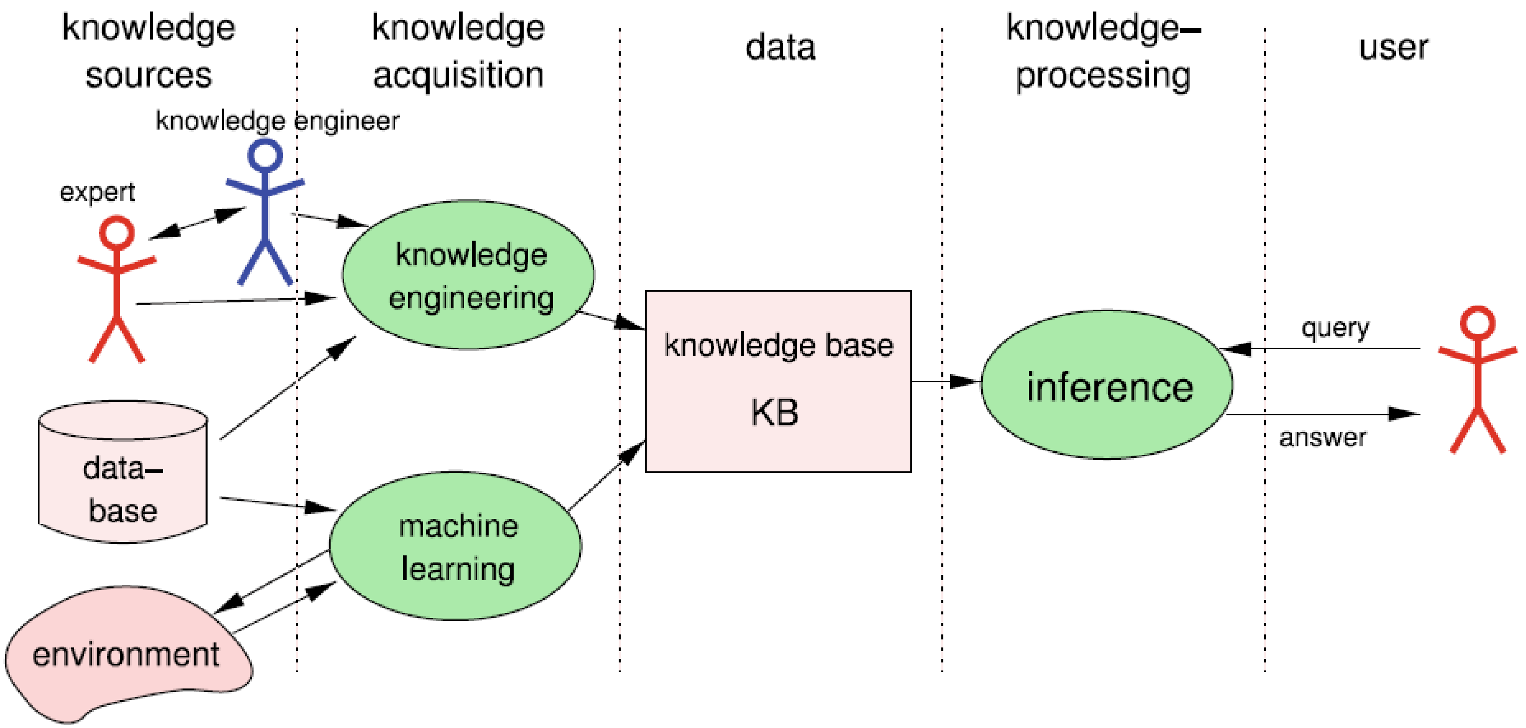


Figure 1.7: Structure of a classic knowledge-processing system

1.5 Knowledge-Based Systems

- Moving toward a *separation* of **knowledge** and **inference** had several crucial advantages.
- The separation of knowledge and inference can allow inference systems to be implemented in a largely *application-independent* way.
- For example, application of a medical expert system to other diseases is much easier by replacing the knowledge base rather than by programming a whole new system.
- Through the decoupling of the knowledge base from inference, knowledge can be stored **declaratively**. In the knowledge base there is only a *description* of the knowledge, which is independent from the inference system in use.

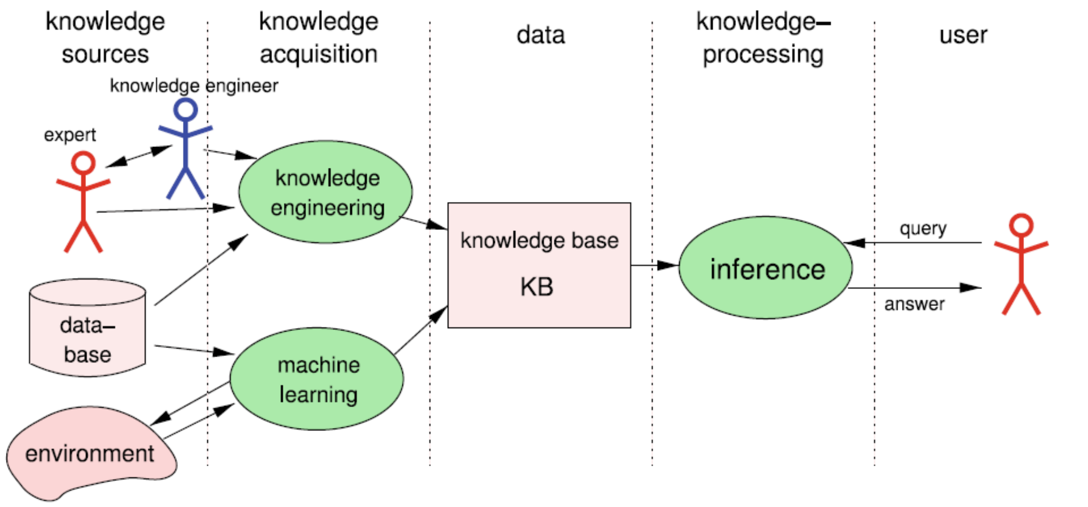


Figure 1.7: Structure of a classic knowledge-processing system

1.5 Knowledge-Based Systems

- **Formal languages** as a convenient interface between man and machine lends itself to the representation of the knowledge base.
- A few examples are *propositional calculus*, *first-order predicate logic* (PL1), *probabilistic logic* and *decision trees*. [**URL**: Propositielogica, Predicatenlogica , Probabilistic Logic, Beslissingsboom (Decision Tree)]
- As an example for a large scale knowledge based system we want to refer to the software agent "**Watson**".
- Developed at **IBM** together with a number of universities, **Watson** is a question answering program, that can be fed with clues given in natural language. It works on a **knowledge base** comprising *four terabytes* of hard disk storage, including the full text of Wikipedia.
- Watson was developed within IBM's **DeepQA project** which is characterized as follows:

*The DeepQA project at IBM shapes a grand challenge in Computer Science that aims to illustrate how the wide and growing accessibility of natural language content and the integration and of advancement of **Natural Language Processing**, **Information Retrieval**, **Machine Learning**, **Knowledge Representation and Reasoning** and **massive parallel computation** can drive open-domain automatic Question Answering technology to a point where it clearly and consistently rivals the best human performance.*

1.5 Knowledge-Based Systems

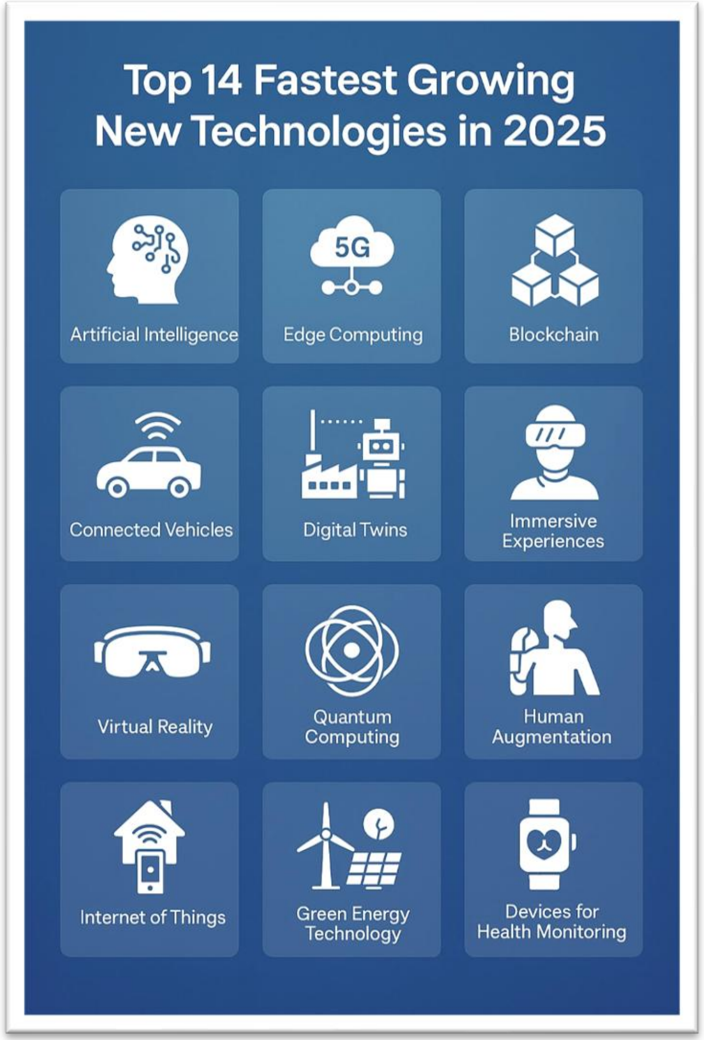
- In the U.S. television quiz show “**Jeopardy**!”, in February **2011**, **Watson** defeated the two human champions **Brad Rutter** and **Ken Jennings** in a two-game, combined-point match and won the *one million dollar price*.

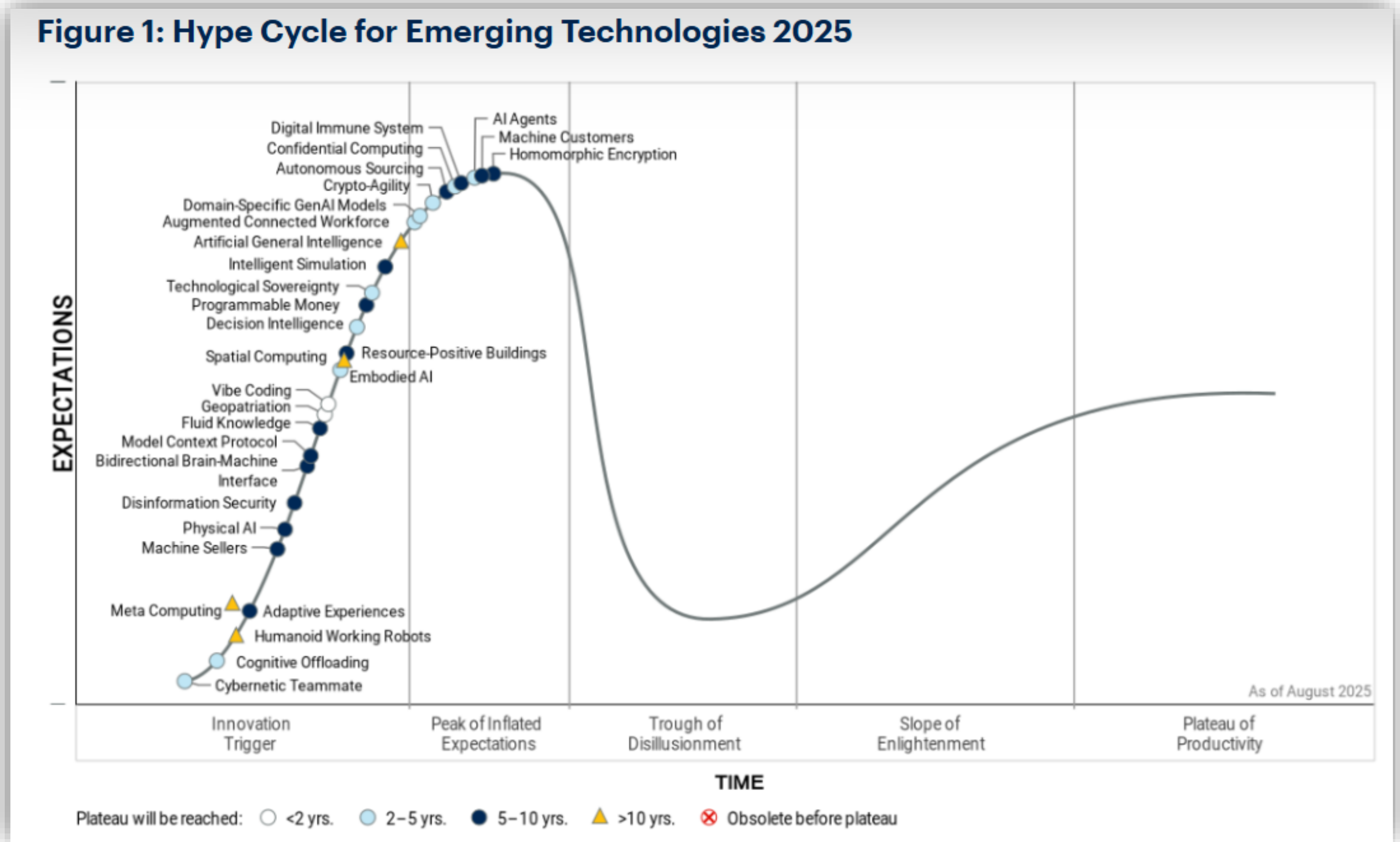
- One of Watson’s particular strengths was its **very fast reaction** to the questions with the result that Watson often hit the buzzer faster than its human competitors and then was able to give the first answer to the question.

- The **high performance** and **short reaction times** of Watson were due to an implementation on **90 IBM Power 750 servers**, each of which contains 32 processors, resulting in **2880 parallel processors**.

Chapter 1: Introduction – Hype Cycle for Emerging Technologies

What about the (current) hype?

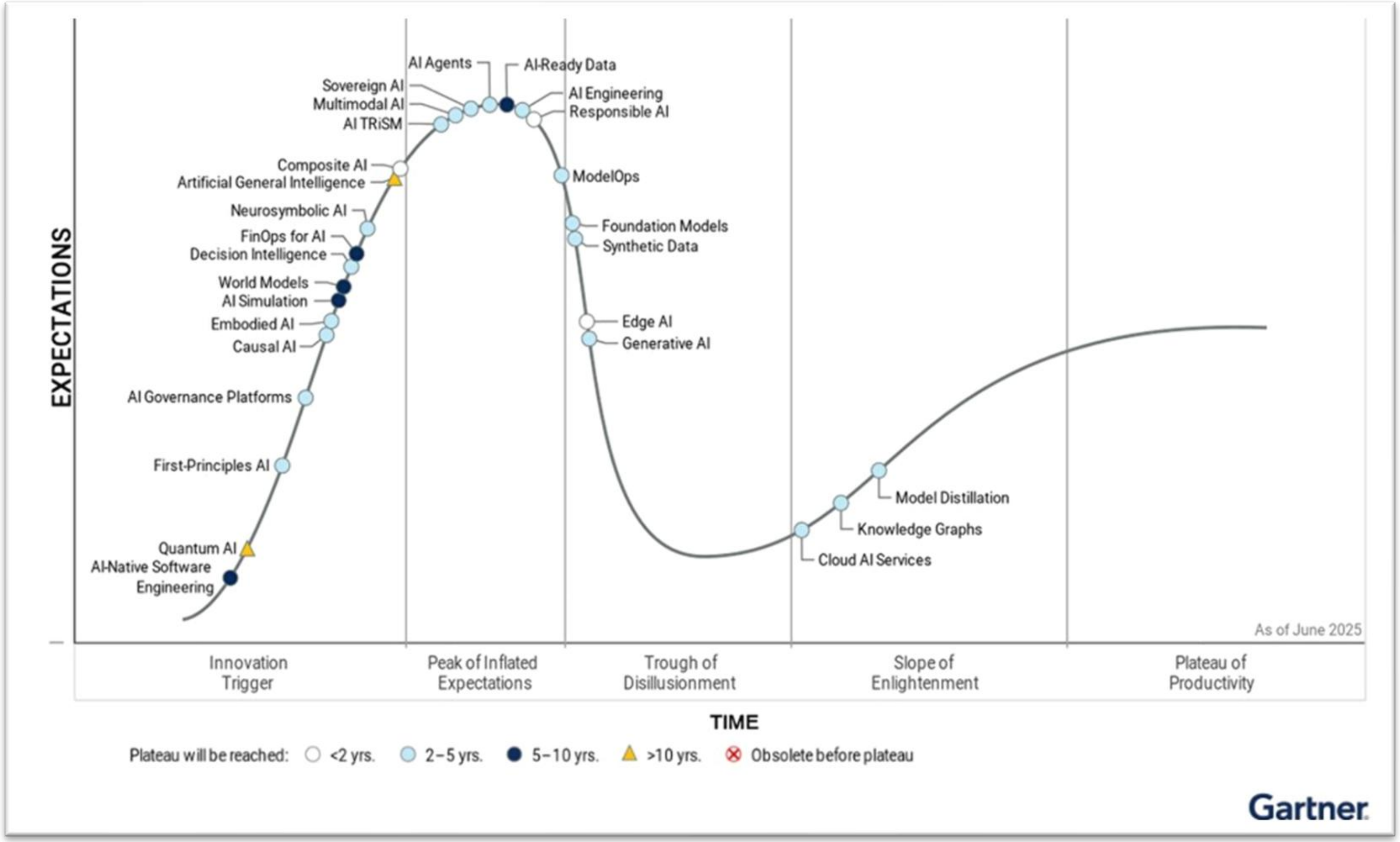




URL: Hype Cycle for Emerging Technologies 2025

Chapter 1: Introduction – Hype Cycle for Artificial Intelligence

What about the (current) hype?



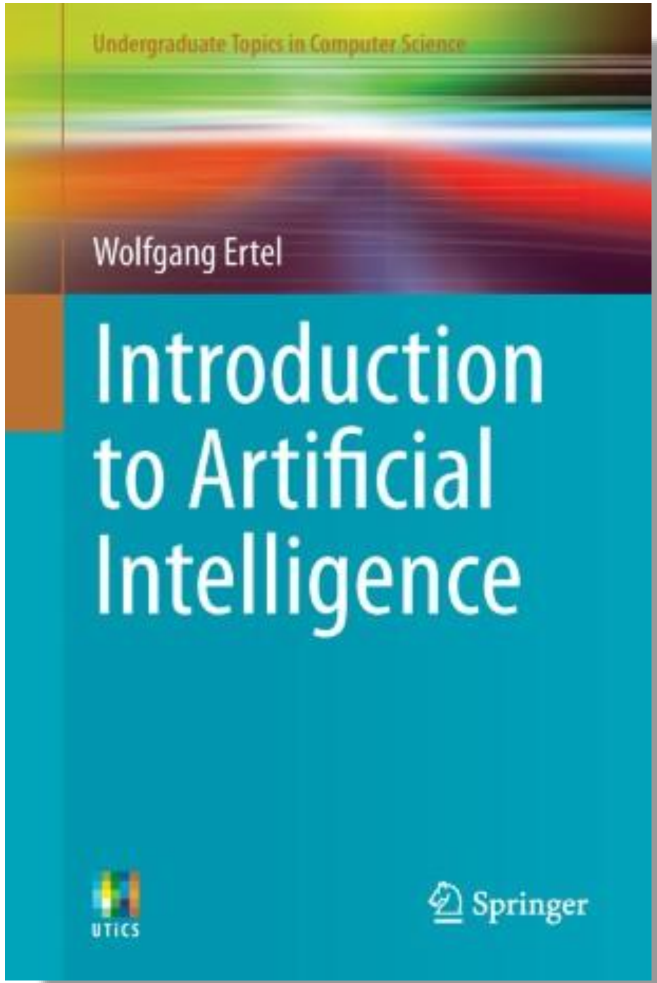
URL: [Gartner Hype Cycle - Top AI Innovations in 2025]

In this introductory course…

- Some **history** on AI
- Some information on **Machine Learning** and **Deep Learning**
- **Basics of AI**
  - *Modelling* of typical problems
    - E.g. chess
  - *Searching* for “good” solutions to probems
    - Basic search
    - Blind search
    - Heuristic search

AI Fundamentals

End of Chapter 1- Introduction and Examples



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